



SKILL BASED PROFILE MAPPING SYSTEM FOR PROJECT ASSISTANCE USING MACHINE LEARNING

**P.H.V.Sesha Talpa Sai^{1*}, Yetukuri Gouthami Maheswari², Nareddy Neha Sri²,
Abhinaya Polneni², Singi Bhanu Sri²,**

¹ Director, Research & Development, Malla Reddy University, Hyderabad, India.

²Department of Information Technology, Malla Reddy Engineering College for Women, Hyderabad, India.

Abstract : By 2030, 64% of India's population will be in the working age range, making it the youngest nation in the world. To improve the employability of the young labor force, the Developing skills and generating high-value jobs in the primary and secondary sectors are the main objectives of the Twelfth Plan strategy. Candidates often face difficulties in finding people with specific skills, such as engineers who struggle to team up with people who can complete project requirements. To address this, a profile suggestion system is proposed that considers a project and suggests people with relevant skills. The system uses machine learning to make accurate predictions based on similarities between project requirements and a person's skills. This tool helps candidates find assistance for their projects and knowledge about skills with more demand in the market. It also allows people looking for freelancing opportunities to apply their expertise knowledge on a project. The profile recommendation system must adapt to novel algorithms and techniques to achieve better job and skill recommendations. Hybrid filtering, a combination of content and collaborative filtering, can solve challenges in content and collaborative filtering, enhancing classification accuracy and achieving better job and skill recommendations.

IndexTerms - Profile Mapping, Machine learning, Skills, Project assistance, Freelancing

I. INTRODUCTION

Recommendations are employed in a variety of industries, such as media, retail, and journalism, and they are a major source of income for many companies. Large volumes of data about consumer behavior are available, allowing you to use machine learning to make extremely relevant recommendations. One of the primary ways for people to locate work is through online job boards or recruiting websites like Indeed or LinkedIn. Both job searchers and hiring companies have benefited greatly from the time and money these online platforms have been saving for over a decade. Apart from that these platforms have also helped in finding people who showcase their primary skills and seek opportunities for freelancing. Candidates who really want the engineers who can assist them in project go across many profiles but fail to find the relevant profiles because of inappropriate information posted by users. The one more cause of these irrelevant profile mapping is because the social media handle's main objective is not to serve this purpose, because of this Niche requirement, often we won't find people who can really help is completing project.

To overcome the problems from both perspectives we have developed a tool using Logistic Regression model which will be breaking down the project into various domains and prepare an overview of what all are the technical domains involved and what skills are required to complete the project. After that this project will be posted on the tool by the user and along with that keywords will be also attached which will be demonstrating that which skilled persons the user is looking for? The person who is actively seeking such chances will see this initiative, map out their skills, and get in touch.

Also when the user who has posted the project goes to search profile by skills tab and search for persons with required domain knowledge, the recommendation model will fetch the profiles from the database and suggest candidates. Now the user can directly get in touch with them and explain them the requirement. The recommendation system presents user-interesting data based on previously acquired information about the user. The goal of personalized results is to give the user pertinent information based on their basic data or system activities. Ontology can be used to model a user's fundamental information into a user profile. The systematic representation of different items and their interactions within a domain is known as ontology. The system creates a user profile by gathering basic data. The algorithm then uses the dynamic features, including the list of favorite jobs and recently seen jobs, as a source of data. In order to provide the user with a list of pertinent jobs, the recommendation algorithm processes the input.

2. LITERATURE SURVEY

Collaborative filtering is one of the best and most suitable methods for making recommendations. The recommendation system's primary objective is to forecast the different products that a user would be interested in based on their preferences.

In the past, user-based collaborative filtering techniques have proven to be highly effective in recommending things based on user preferences. However, when the number of people and goods rises, there are also some issues that need to be addressed, like scalability and sparse data. It can be challenging to locate the relevant information quickly on a huge website. However, information and products that are most appropriate for us are filtered out by the recommendation system. Despite the existence of other suggestion methods, the collaborative filtering methodology is highly favored due to its efficacy. This article describes a movie recommender system that essentially employs collaborative filtering, an item-based technique, to generate recommendations for items. The system is dynamic and learns from positive input of the recommender system's history [2]. This study proposes a job recommender system based on dynamic user profiles. To address the problem of job applicants not updating their profiles on time, we dynamically update and expand the user profile based on the applicant's past applied jobs and behaviors. In order to enhance conventional IR methods, a number of published studies take into account talents as job description keywords that must be matched in order to identify the optimal job recommendation. They employ both content-based and collaborative filtering-based approaches to construct recommendations, taking into account user-job dyadic data. Skill is not an active element in these models; instead, it is utilized subtly as a keyword in the user and job profiles. Using sequences of many users' work histories, generative models have also been used to perform job recommendations. Such a model ignores abilities related to a particular job and solely uses career history data [3, 4].

Most of the suggested works view skills as passive components of the job description that must be matched in order to determine the most appropriate job proposal. A few published papers introduced content-based filtering [13]. However, the ramp-up issue with CBF approaches is that they need to gather enough ratings in order to build a trustworthy classifier. Furthermore, they are constrained by the attributes that are specifically associated with the items they suggest. Collaborative filtering was introduced by others [14] to take interests of similar users into account but it cannot handle fresh items also known as the ‘Cold Start’ problem. The Big Five personality test was introduced by few other authors [15] to make the recommendation personalized but it is time consuming to generate a profile for each user. Some other authors used Skill Extraction, Vectorization, Clustering, Comparison and Analysis to filter CVs of applicants [16]. But it has limited novelty in that it cannot personalize the results. Kumar et.al. calculated skill value and predicted salary based on those skill values. They found that the higher the value for a skill, the higher the job is paid [17].

Employee performance and skill level are evaluated in manufacturing businesses using a variety of assessment instruments, including the skill charting method and performance appraisal. One method for evaluating employees' skill ratings is skill mapping. Based on skill mapping, employees are guided to learn new skills or areas for growth if necessary. This will increase employee engagement and, consequently, employee performance. But our approach is to implement this skill mapping for project assistance for final year graduates and post graduates. For this project, we have used a Collaborative filtering approach, combined with Content-based filtering and Hybrid Recommendation Systems. To integrate this, we have used front end technology and backend technology. Fig. 1 shows the landing page of the application where the user can head over to the service.



3.1 Front-end Development

3.2 Backend Development with Python Flask and MySQL

With Flask Libraries, developers can quickly and simply create lightweight web apps using the Flask web framework. We used Flask for our application because of its simplicity and easy integration with web pages designed using HTML and CSS. The language we use to write the code is Python. Basic instructions like connecting to database, redirecting to other web pages and several functions are written in Python language.

The relational database management system (RDBMS) MySQL is open-source. To help structure the data, a relational database arranges the data into one or more tables where the data may be related to one another. Programmers can build, edit, and retrieve data from relational databases using SQL, which also allows them to manage user access. We have used MySQL for our data collection and retrieval.

3.3 Recommendation Model

Recommender systems employ a technique called collaborative filtering (CF). There are two aspects of collaborative filtering: a more broad one and a more specific one. Collaborative filtering, in its more recent and limited definition, is a technique that gathers preferences or taste data from numerous users in order to automatically forecast (filter) a user's interests. The practice of filtering information or patterns using methods that involve cooperation amongst several agents, points of view, data sources, etc. is known as collaborative filtering in a broader sense. We have used CF to train it on past data i.e., users with similar skills sets or projects are clustered. CF helps in efficiently finding similarities how different users work on projects, suggesting people with relevant skills. Fig.2 shows the basic principle behind CF.

The standard procedure for a collaborative filtering system is:

1. To express their choices, a user ranks system items (such books, movies, or music recordings). These ratings can be seen as a rough representation of the user's degree of interest in the subject at hand.
2. By comparing this user's ratings to those of other users, the system determines who has the most "similar" tastes.
3. Products that comparable users have rated highly but that this user has not yet rated are suggested by the algorithm (apparently, the absence of a rating is usually seen as the unfamiliarity of an item).

The method of combining and weighing user neighbors' choices is a major challenge in collaborative filtering. Users can occasionally rate the suggested things right away. Consequently, the system gradually acquires a more precise depiction of human preferences.

Another kind of collaborative filtering might be based on implicit observations of usual user behavior, as opposed to the artificial behavior imposed by a rating task. In order to predict a user's future behavior or how they might prefer to behave if given the chance, these systems consider both the user's past actions and those of other users (e.g., what music they have listened to or what they have bought). To ascertain how these predictions might impact a business system's operations, they must thereafter be filtered by business logic. Offering to sell someone a certain album of music, for instance, is pointless if they have already proven they are the owner of that album.

Using a rating or scoring system that averages scores from all users overlooks user preferences and performs poorly on tasks where interest varies greatly (e.g., music suggestion). Other strategies, like web search and data clustering, can be used to counteract the information explosion.

Based on Content on the user's past behavior or explicit feedback, filtering uses item attributes to suggest other items that are similar to what they like. Items that are pertinent to this user should be suggested by the model. You must first choose a similarity metric (such the dot product) in order to accomplish this. The system must then be configured to assign a score to each potential item based on this similarity metric. Fig 3 demonstrates how the content-based filtering works. The information that we gather from the user—either explicitly (by rating) or implicitly (by clicking on a link)—is what a content-based recommender uses to function. We use the data to build a user profile, which is then used to provide recommendations to the user. The engine gets more accurate as the user adds additional information or acts on the advice.

In order to forecast user-item interaction and make appropriate item recommendations, the recommendation engine in CBRs basically matches an item profile with a user profile.

- An object's item profile is how the system represents it. It consists of an item's feature set, which could comprise internal structural characteristics or descriptive metadata. For instance, movies can be arranged by director, genre, and release date on a streaming service.
- A user's profile shows their conduct and interests. It might contain illustrations of the items in which a user has previously shown interest. It also contains user data from past encounters with the system, including queries, ratings, likes, dislikes, and so on.

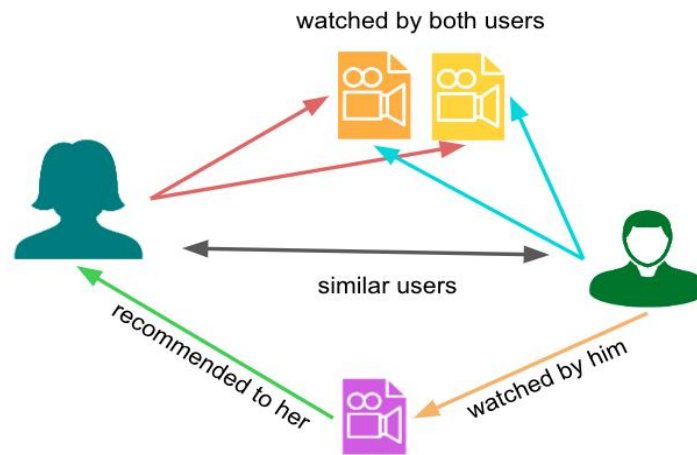


Fig. 2 CF basic principle

The cold-start problem essentially refers to how a system reacts to new users or items. Both present difficulties since collaborative filtering groups users according to assumed behavioral and preference similarities. Nevertheless, there is no proof that new users are similar to existing ones, and user participation (such as ratings) is too low to support the recommendation of new items. Content-based filtering does a good job at introducing new items, but it struggles with new users. This is because it makes product recommendations based on internal characteristics or metadata rather than past user encounters.

We have integrated these two approaches and formed a Hybrid model. This model was suitable for our project, as we were able to able to recommend projects based on similarities between users and also recommend projects based on the interest of sole user.

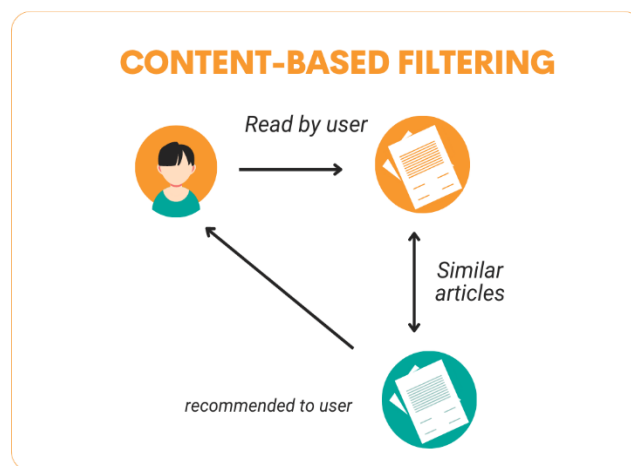


Fig. 3 Content based filtering approach

4. SYSTEM ARCHITECTURE

As seen in Fig 6. The flow of this application is simple and precise. The user who is looking for people who can help him/her in completing a project will register using the Post A Project option from the application. The applicant will be prompted to enter basic details like Name, E mail, Contact Number, etc. This will be followed by the next registration form where the user must enter the project details, The project details will be including Project title, abstract, technologies involved, number of team members required and finally the important aspect: - skill required for completing the project. The registration form of Post a project option looks something like Fig. 4.

On the other hand, the person looking for project will also be prompted to complete the basic registration such as Name, E mail, Contact number and so on. The next step will be the user searching for a project in the particular domain he/she has expertise with as shown in Fig 5. The project posted by the person in case 1 will be mapped with this user and the details will be shared. The person who has posted the project will be receiving this person's info and can contact him/her directly and discuss the proceedings.

Fig. 4 Post a project

Fig 5. Get a Project snapshot

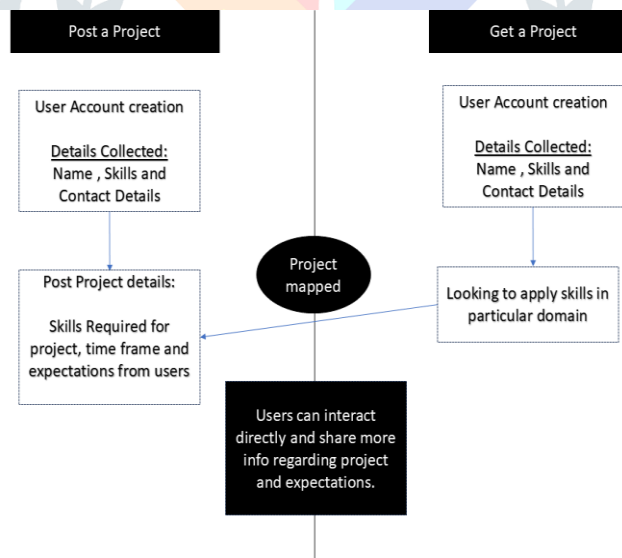


Fig 6. Application work flow

5. RESULT & DISCUSSION

This application was tested by deploying onto Netlify and the application work flow is shown in Fig. 6. The model in the background was working absolutely fine and was recommending projects based on skills as expected. The important factor that we were able to achieve was not only the exact keyword was matched, but the whole technology was being mapped for the project. For example, if a person has posted a project based on Web Development and is looking for a person's help in React technology, The system was successfully suggesting this project to all the persons who have mentioned Front End Development in their areas of skills, because React comes under the domain of Front End Development. As demonstrated in Fig.7, the system was matching the project and showing the list of all available projects to users looking for them. As we can see in the Fig 7, the application has successfully recommended 2 projects posted by people to a user who is looking to apply his/her skills in the field of Python.

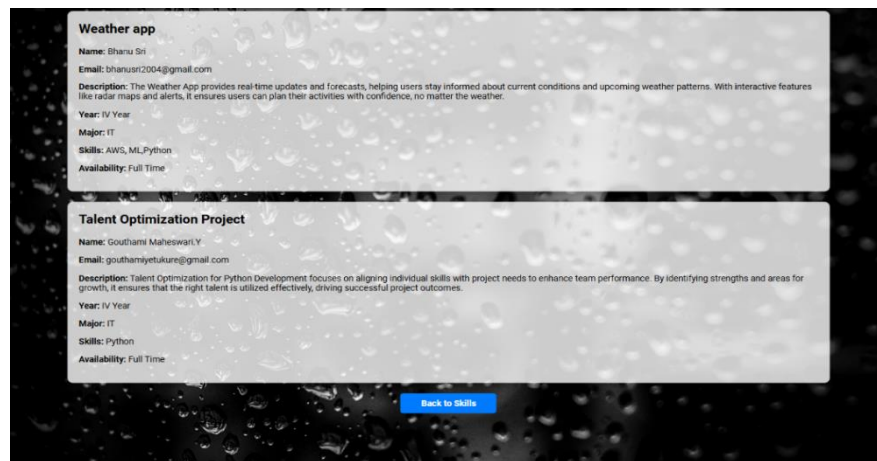


Fig. 7. Skill Mapping based on Project

6. REFERENCES

- [1] R. J. Mooney and L. Roy, "Content-Based Book Recommending Using Learning for Text Categorization," in In Proceedings of DL '00: Proceedings of the Fifth ACM Conference on Digital Libraries, New York, NY, pp. 13-20, 2000
- [2] Li-Ping Jing, Hou-Kaun Huang, Hong-Bo Shi, "Improved feature selection approach TFIDF in text mining", International Conference on Machine Learning and Cybernetics, pp. 944-946, 2002, doi:10.1109/icmlc.2002.1174522.
- [3] Lu, Y.; El Helou, S.; Gillet, D. A Recommender System for Job Seeking and Recruiting Website. In Proceedings of the 22nd International Conference on WWW. Association for Computing Machinery, Rio de Janeiro, Brazil, 13–17 May 2013; pp. 963–966.
- [4] Xu, Y.; Li, Z.; Gupta, A.; Bugdayci, A.; Bhasin, A. Modeling Professional Similarity by Mining Professional Career Trajectories. In Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, New York, NY, USA, 24–27 August 2014; pp. 1945–1954.
- [5] Junior, J.; Vilasbôas, F.; De Castro, L. The Influence of Feature Selection on Job Clustering for an E-recruitment Recommender System. In Proceedings of the International Conference on Artificial Intelligence and Soft Computing, Zakopane, Poland, 12–14 October 2020; Springer: Cham, Switzerland, 2020; pp. 176–187.]
- [6] Mohammadzadeh, H.; Gharehchopogh, F.S. An efficient binary chaotic symbiotic organisms search algorithm approaches for feature selection problems. *J. Supercomput.* **2021**, *77*, 9102–9144.
- [7] Mohammadzadeh, H.; Gharehchopogh, F.S. Feature Selection with Binary Symbiotic Organisms Search Algorithm for Email Spam Detection. *Int. J. Inf. Technol. Decis. Mak.* **2021**, *20*, 469–515.
- [8] Abdollahzadeh, B.; Gharehchopogh, F.S. A Multi-Objective Optimization Algorithm for Feature Selection Problems. *Eng. Comput.* **2022**, *38*, 1845–1863.
- [9] Naseri, T.S.; Gharehchopogh, F.S. A Feature Selection Based on the Farmland Fertility Algorithm for Improved Intrusion Detection Systems. *J. Netw. Syst. Manag.* **2022**, *30*, 40.
- [10] Goldanloo, M.J.; Gharehchopogh, F.S. A hybrid OBL-based firefly algorithm with symbiotic organisms search algorithm for solving continuous optimization problems. *J. Supercomput.* **2022**, *78*, 3998–4031.
- [11] Ntioudis, D.; Masa, P.; Karakostas, A.; Meditskos, G.; Vrochidis, S.; Kompatsiaris, I. Ontology-Based Personalized Job Recommendation Framework for Migrants and Refugees. *Big Data Cogn. Comput.* **2022**, *6*, 120.
- [12] Mishra, R.; Rathi, S. Enhanced DSSM (Deep Semantic Structure Modelling) Technique for Job Recommendation. *J. King Saud Univ. Comput. Inf. Sci.* **2022**, *34*, 7790–7802.
- [13] Shaha T Al-Otaibi and Mourad Ykhlef [2012] - A Survey Of Job Recommender Systems. International Journal of Physical Sciences , Vol. 7, 29 (2012), 5127--5142.
- [14] M Viswa Murali, T G Vishnu, Nancy Victor [2019] - A Collaborative Filtering based Recommender System for Suggesting New Trends in Any Domain of Research.
- [15] Orestis Nalmpantis & Christos Tjortjis [2017] - The 50/50 Recommender: A Method Incorporating Personality into Movie Recommender System.
- [16] Koen Bothmer and Tim Schlippe [2022] -Skill Scanner: Connecting and Supporting Employers, Job Seekers and Educational Institutions with an AI-based Recommendation System.
- [17] Naresh Kumar, Manish Gupta, Deepak Sharma and Isaac Ofori [2022] - Technical Job Recommendation System Using APIs and Web Crawling